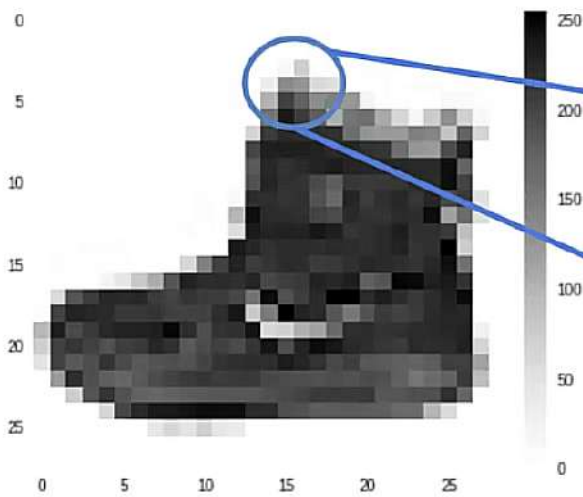




0	64	128
48	192	144
142	226	168

Current Pixel Value is 192

Consider neighbor Values

-1	0	-2
.5	4.5	-1.5
1.5	2	-3

Filter Definition

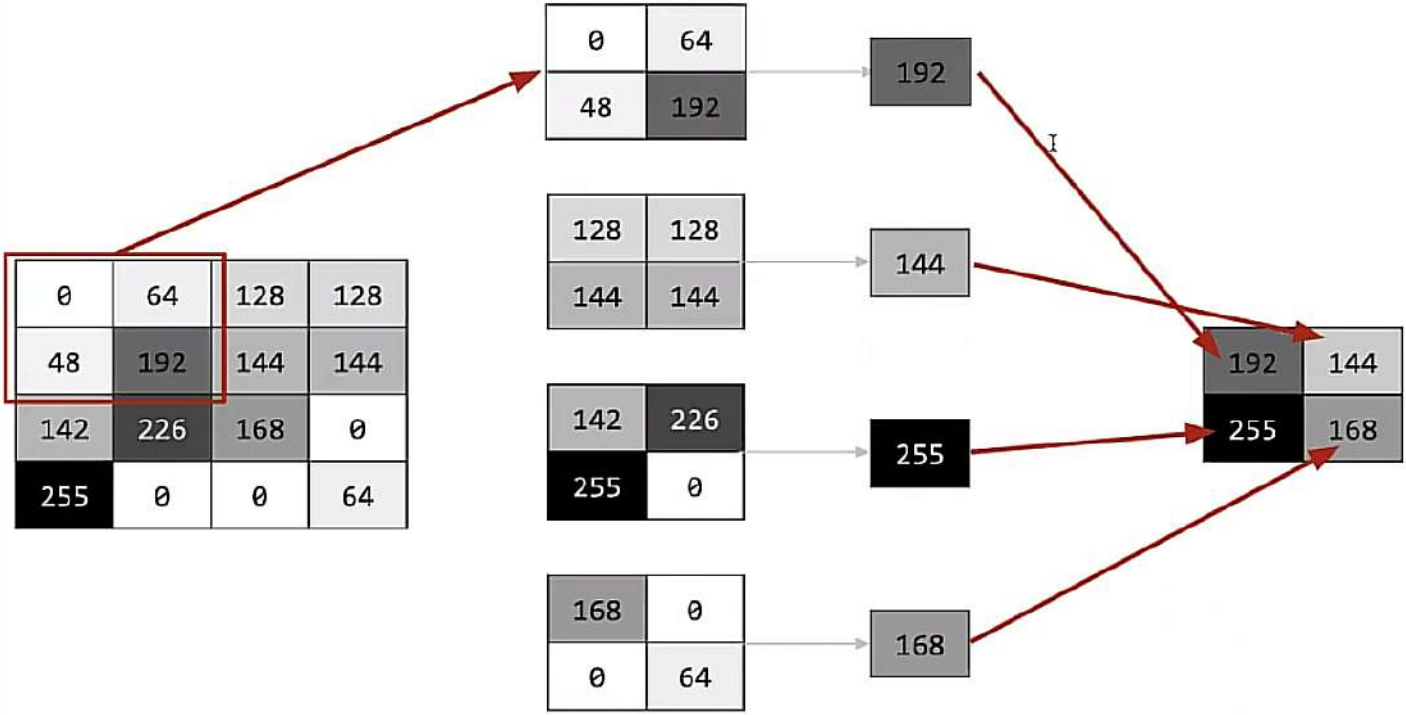
CURRENT_PIXEL_VALUE = 192

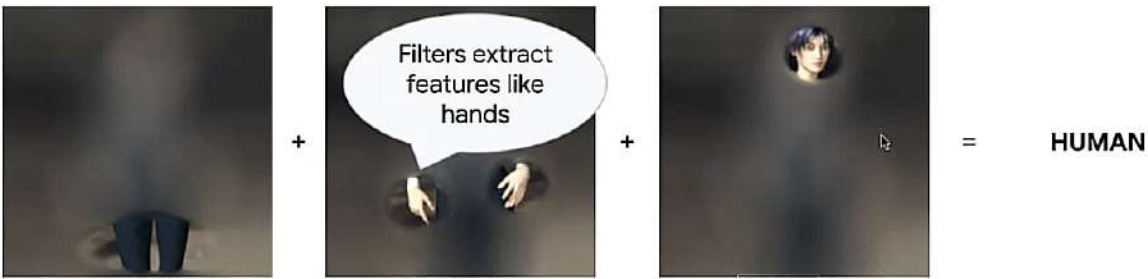
NEW_PIXEL_VALUE = $(-1 * 0) + (0 * 64) + (-2 * 128) +$
 $(.5 * 48) + (4.5 * 192) + (-1.5 * 144) +$
 $(1.5 * 42) + (2 * 226) + (-3 * 168)$



-1	0	1
-2	0	2
-1	0	1

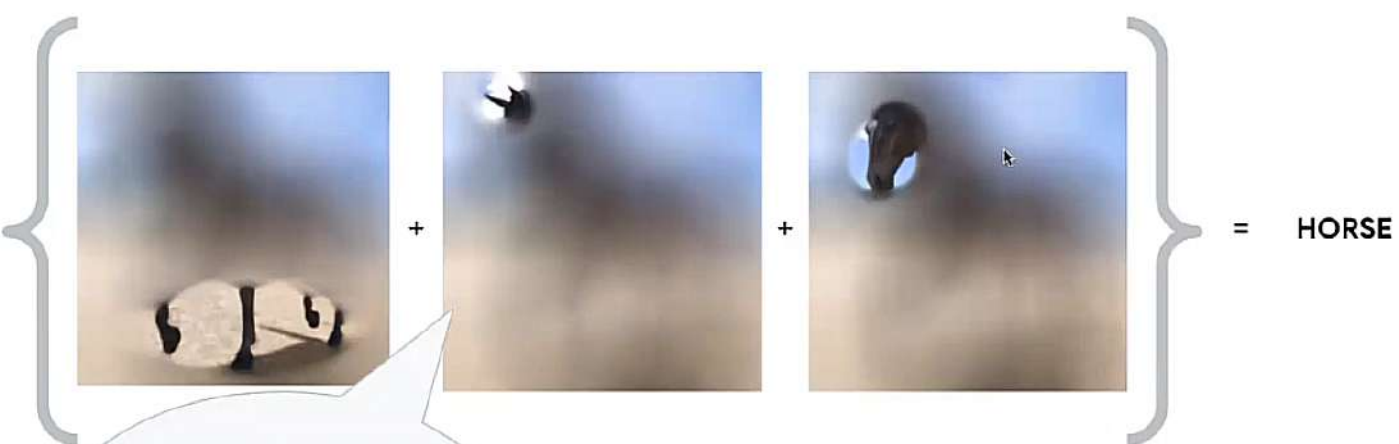






Rahul Mangharam: So, over time, in the same way as the activity recognition earlier if the computer can figure out which filters extract the features that determine the difference between the also have a human or the answer for horse to recap for does extract features like happens for years.



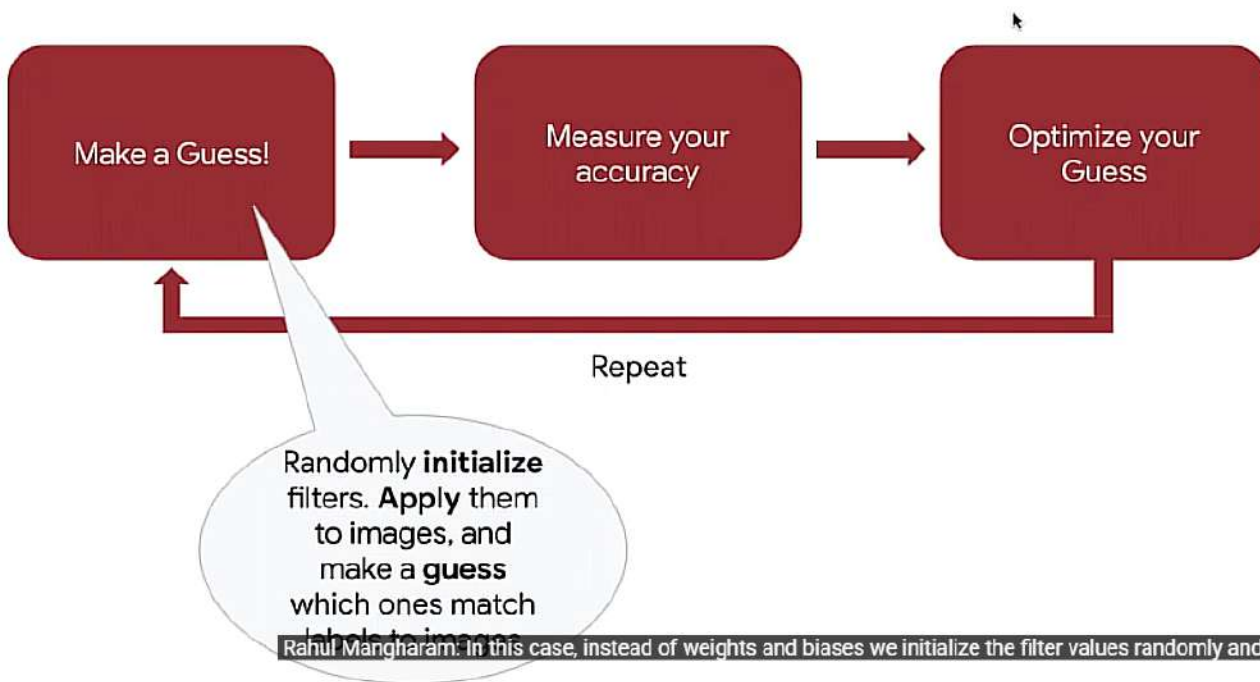


Filters can then be combined with **labels** to make a **prediction** of the image contents...

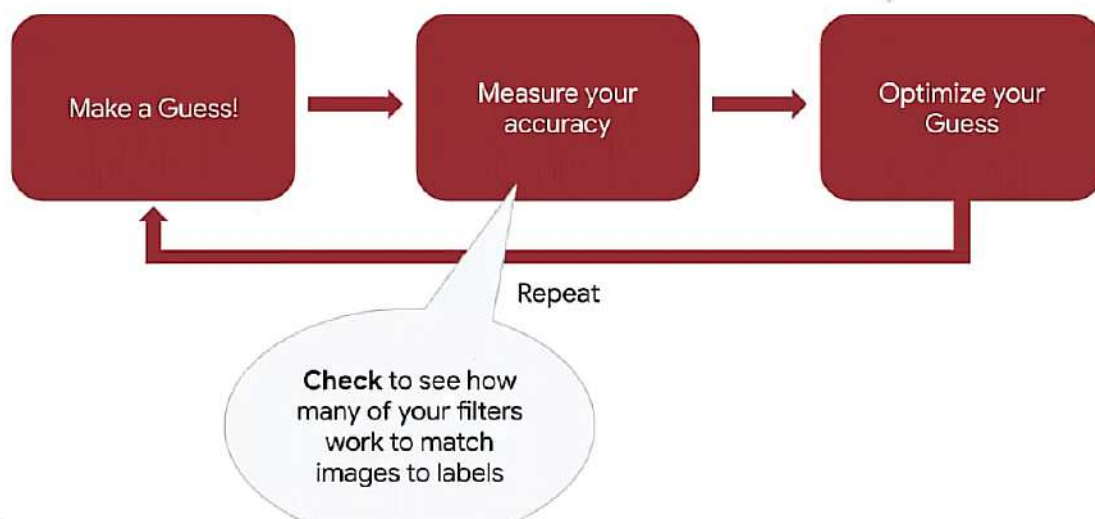
Rahul Mangharam: And during training can be combined with the labels of the known images to help us train a network that can predict which parts.



The Machine Learning Paradigm

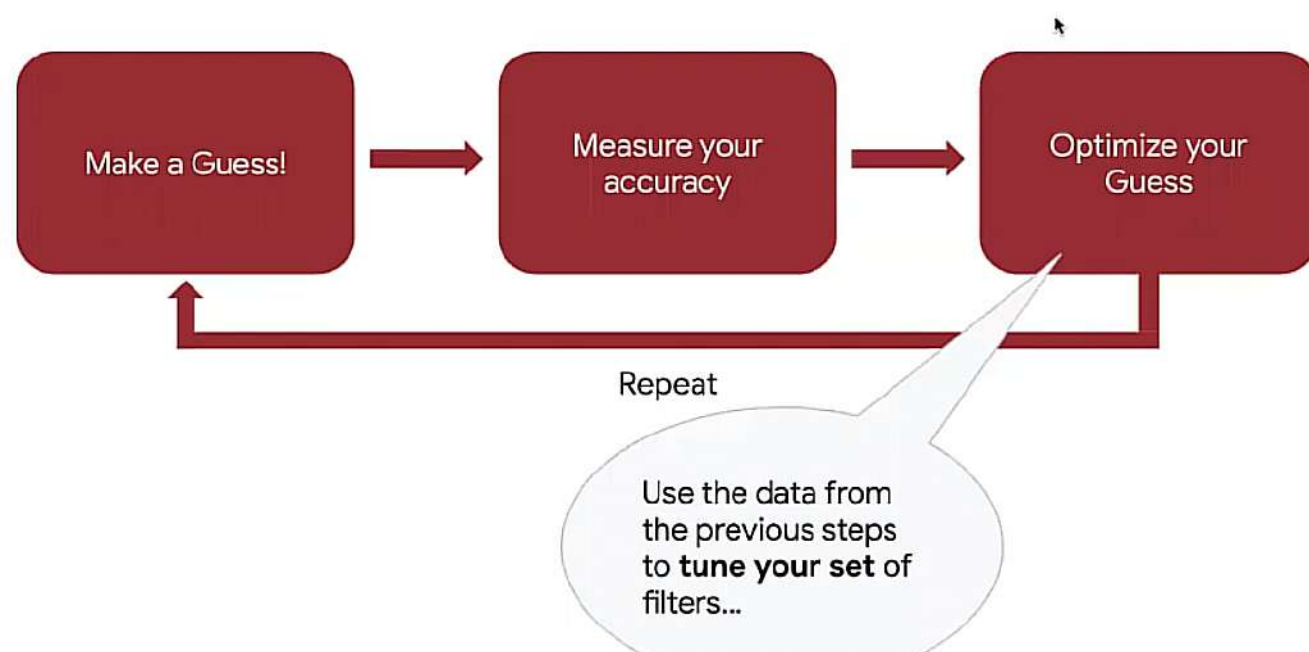


The Machine Learning Paradigm



Rahul Mangharam: which give us our guesses by mentioning our guesses as to the class of the image, based on the extracted features against the ground truth, we can measure the accuracy. ...

The Machine Learning Paradigm



Rahul Mangharam: And as before we can use an optimizer to tweak our filter values, knowing that will be stepping in the right direction.